

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I P. Karthik - 192524822 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

CO-1 Assessment tool

-1

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I) Smart home automation system:

A smart home automation system is a home that uses Artificial intelligence, sensors, and smart devices to automatically control appliances like fans, lights, AC, doors etc.

Main Objectives:

- Improve comfort.
- Save electricity.
- Increase home security.
- Learn user preferences.

Types of Intelligent Agents:

a) Simple reflex agent

- Works using condition-action rules.
- Ex: If motion is detected → Turn on lights.
- Advantage: Fast and simple.
- Disadvantage: Cannot learn or remember past events.

6) Model-Based agent:

- Maintains an internal model of the home.
- Ex: Keeps track of room temperature even when sensors are temporarily inactive.
- Advantage: Works better in changing environments.

c) Goal-Based Agent

- Makes decisions to achieve a specific goal.
- Ex: Maintain room temperature even when sensors ensuring home security.

d) Utility-Based Agent:

- Chooses the action with the highest overall benefit.
- Ex: Balances comfort, energy saving and security.
- Advantage: Gives the best-overall performance.

Best choice:

A hybrid agent is the most effective because it:

- learns user preferences.
- Achieves comfort and security goals.
- Saves energy by selecting the best action.

2) Robot in an Unknown Maze:

A Robot in an unknown Maze is an AI problem where a robot is placed inside a maze without knowing its layout. The robot must find the exit by exploring different paths.

Since the robot has no map, it must use search algorithms to decide which path to take.

Main Objectives

- Explore unknown paths.
- Avoid obstacles and dead ends.
- Find the exit successfully.
- Choose the shortest or least-cost path.

Uniformed Search Strategies

a) Uniform Cost search (UCS)

- Expands the path with the lowest cost first.
- Advantages:
 - Finds the optimal (lowest-cost) path.

- Complete if costs are positive.
- Disadvantages:
 - Uses more memory.
 - Slower for large mazes.

6) Iterative Deepening Search (IDS)

- Repeatedly performs depth limited search with increasing depth.

• Advantages:

- Requires very little memory.
- Complete for finite search spaces.

• Disadvantages:

- Repeats some searches, increasing execution time.

Relation to Intelligent Agents :

- Simple reflex agent: Reacts only to current position, poor for maze solving.

- Model-Based Agent: Remembers explored paths and avoids revisiting.

- Goal Based Agent: Plans actions to reach the exit.

- Utility Based Agent: Chooses the safest or least-cost path.